

Measuring CEO Individualism from Earnings Calls: Construction, Validation, and Initial Findings

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Abstract

We develop a large language model (LLM)-based measure of CEO individualism from 463,646 earnings call transcript segments covering 3,480 CEOs at 1,601 firms over 2005–2024. The measure captures whether a CEO presents themselves as an individual leader (“I decided”) versus an organizational representative (“we as a team decided”) along three dimensions: self-referential leadership, personal communication style, and organizational embedding. We validate the measure extensively: split-half reliability is 0.93 (Spearman-Brown corrected), convergent validity with first-person pronoun ratio is $r = 0.62$, the measure is orthogonal to overconfidence ($r = -0.07$), and only 5.6% of its variance is explained by observable CEO and firm characteristics. We document several findings. First, CEO individualism is associated with higher analyst coverage, lower bid-ask spreads, and lower illiquidity, but not with improved forecast accuracy—a pattern we term “attention without information.” Second, the interaction between individualism and abnormal stock returns significantly predicts forced CEO turnover in the following year ($z = -3.67$, $p = 0.0002$ in logit): individualist CEOs face lower turnover after good performance but lose this protection after poor performance. Third, individualism is cross-sectionally associated with lower R&D intensity and higher capital expenditure. Fourth, average CEO individualism has declined steadily over the past 20 years, driven entirely by cohort replacement rather than within-CEO change.

Keywords: CEO individualism, earnings calls, LLM, forced turnover, analyst coverage, corporate governance

1 Introduction

A growing body of research documents that individual CEO characteristics—overconfidence (Malmendier and Tate, 2005), narcissism (Chatterjee and Hambrick, 2007), military experience (Benmelech and Frydman, 2015), general versus specialist background (Custódio et al., 2013)—influence corporate policies and outcomes (Bertrand and Schoar, 2003). In parallel, the textual analysis literature has shown that the language of corporate disclosures contains economically relevant information (Loughran and McDonald, 2011). However, these two literatures have largely developed independently: the CEO characteristics literature relies on indirect proxies (option exercise, photograph size, biographical records), while the textual analysis literature focuses on *what* is said rather than *who* is saying it and how.

This paper connects these two strands by developing a new measure of CEO personality—individualism—directly from how CEOs speak during quarterly earnings calls. Individualism–collectivism is one of the most extensively studied dimensions in psychology (Hofstede, 1980; Markus and Kitayama, 1991; Triandis, 1995). At the individual level, it captures whether a person construes the self as independent (“I am the decision-maker”) or interdependent (“We as a team make decisions”). We operationalize this construct along three dimensions scored from CEO speech: self-referential leadership, personal communication style, and organizational embedding.

We construct the measure using a two-stage LLM pipeline. First, Claude Sonnet (Anthropic) scores 20,000 randomly sampled earnings call segments along the three dimensions using a detailed rubric. Second, we fine-tune an open-source Qwen3-14B model on these labels ($R^2 = 0.75$ on held-out test set) and deploy it to score the full corpus of 463,646 CEO speech segments. The resulting dataset covers 3,480 CEOs at 1,601 firms from 2005 to 2024.

We validate the measure along multiple dimensions. Split-half reliability is 0.93 (Spearman-Brown corrected), indicating excellent internal consistency. The measure converges with the first-person pronoun ratio ($r = 0.62$), a standard text-based narcissism proxy, but diverges in predictive patterns—the two measures enter with opposite

signs in regressions predicting capital expenditure. The measure is orthogonal to managerial overconfidence (Holder67; $r = -0.07$) and to corporate political ideology (PAC CFscore; $r = 0.01$). It survives controls for CEO power proxies (tenure, duality, board independence, Delta). Importantly, observable CEO and firm characteristics jointly explain only 5.6% of the variance in the individualism score—the remaining 94.4% is unexplained, consistent with the measure capturing a genuine personality trait.

We then document several empirical associations. We organize our findings into four groups:

Finding 1: Attention without information. CEO individualism is positively associated with analyst coverage (+5.7% per standard deviation, $t = 4.6$), negatively associated with effective bid-ask spreads ($t = -2.5$) and Amihud illiquidity ($t = -3.5$), but unrelated to forecast dispersion or forecast accuracy. This combination suggests that individualist CEOs attract greater market attention without improving the underlying information environment.

Finding 2: Performance-turnover sensitivity. In models predicting forced CEO turnover in year $t + 1$ using CEO characteristics and firm performance in year t , the interaction of individualism with abnormal stock returns is highly significant (logit $z = -3.67$, $p = 0.0002$). Individualist CEOs face lower forced turnover after good performance (rate of 1.04%) but lose this protection after poor performance (1.32%). Collectivist CEOs' turnover is relatively insensitive to performance (2.11% after good, 1.64% after poor).

Finding 3: Corporate investment. Individualism is negatively associated with R&D intensity ($t = -3.3$) and positively with capital expenditure ($t = 2.1$) in cross-sectional specifications with industry×year fixed effects. These associations do not survive firm fixed effects.

Finding 4: Secular decline. Average CEO individualism has declined by approximately 0.12 standard deviations over the sample period (2005–2024), at a rate of -0.006 per year ($t = -14.1$). Decomposition shows this is driven entirely by cohort replacement—newly appointed CEOs are systematically more collectivist—while

within-CEO individualism is stable.

We also document that CEO individualism is higher for male CEOs, founder CEOs, CEOs with dual CEO-chair roles, CEOs with higher ownership, and at firms with less independent boards. These observable predictors jointly explain only a small fraction of the variation, leaving the vast majority attributable to individual personality.

We note important limitations. The cross-sectional associations (Findings 1 and 3) do not survive firm fixed effects or CEO transition event studies, limiting causal claims. The performance-turnover interaction (Finding 2), while robust to controls and consistent in direction, is identified from cross-sectional variation and may reflect CEO-firm sorting. We present these findings as documented patterns that future research—with stronger identification strategies—can build upon.

2 Related Literature and Construct Definition

2.1 Individualism–Collectivism in Psychology

The distinction between individualism and collectivism is among the most studied dimensions of human psychology. Hofstede (1980) identified individualism–collectivism as a primary axis of cross-cultural variation. Markus and Kitayama (1991) formalized it at the individual level as a difference in *self-construal*: independent self-construal (viewing oneself as autonomous, unique, and self-directed) versus interdependent self-construal (defining oneself through relationships, roles, and group memberships). Triandis (1995) extended this framework to behavioral predictions: individualists prioritize personal goals, take individual credit and blame, and communicate in self-referential terms; collectivists prioritize group harmony, share credit, and communicate in team-oriented terms.

At the national level, individualism has been linked to economic outcomes including innovation (Gorodnichenko and Roland, 2011), cross-border mergers (Ahern et al., 2015), and financial risk-taking (Li et al., 2013). However, individual-level measurement within corporate settings has been limited.

2.2 CEO Characteristics and Corporate Outcomes

Bertrand and Schoar (2003) established that individual managers have distinct “styles” that persist across firms and time, affecting investment, financing, and organizational policies. Subsequent work has identified specific traits that drive these styles:

- *Overconfidence*: Malmendier and Tate (2005) measure overconfidence through option exercise behavior and show it predicts excess investment and acquisition activity. Our individualism measure is orthogonal to overconfidence ($r = -0.07$).
- *Narcissism*: Chatterjee and Hambrick (2007) proxy narcissism using CEO photograph prominence and first-person pronoun usage. Our measure converges with the pronoun proxy ($r = 0.62$) but diverges in predictive patterns.
- *Background characteristics*: Benmelech and Frydman (2015) show that military experience predicts conservative investment; Custódio et al. (2013) link generalist backgrounds to higher pay and more strategic change.
- *Celebrity*: Malmendier and Tate (2009) show that CEOs who receive awards subsequently underperform, suggesting that external attention can distort CEO behavior.

Our contribution is to measure a specific communication trait—the degree of individualist self-presentation in public speech—that is related to but distinct from these prior constructs.

2.3 CEO Turnover and Board Monitoring

The sensitivity of CEO turnover to firm performance is a central topic in corporate governance. Weisbach (1988) shows that boards dominated by outside directors are more likely to dismiss underperforming CEOs. Hermalin and Weisbach (1998) models the endogenous relationship between CEO ability, board independence, and turnover. Parrino et al. (2003) documents that institutional investors vote with their feet around

forced CEO turnover events. **Jenter and Lewellen (2015)** examines CEO preferences in acquisition decisions and the consequences for their career outcomes.

This literature typically treats CEO characteristics as determinants of turnover *directly*—e.g., overconfident CEOs may be more likely to be fired because they make worse decisions. We examine a different channel: whether CEO communication style modulates the *sensitivity* of turnover to performance, potentially by shaping how boards attribute firm outcomes to the CEO personally.

2.4 Construct Definition

We define CEO individualism as the degree to which a CEO’s earnings call speech exhibits:

1. **Self-Referential Leadership (D1):** First-person singular framing; personal ownership of decisions and outcomes.
2. **Personal Communication Style (D2):** Distinctive, personality-revealing language (anecdotes, humor, strong opinions) versus standardized corporate language.
3. **Organizational Embedding (D3, reverse):** Operating as a stand-alone decision-maker versus being embedded in and deferring to the organizational hierarchy.

Each dimension is scored on a 1–5 scale from earnings call text. The composite score is the equal-weighted mean of D1, D2, and D3. The detailed scoring rubric is provided in the Technical Appendix.

3 Data and Methodology

3.1 Sample Construction

Our analysis combines data from several sources. Earnings call transcripts are obtained from S&P Capital IQ, covering quarterly conference calls for U.S. public firms from 2005 to 2024. CEO biographical information (identity, tenure, age, gender) comes from

ExecuComp. Firm financial characteristics are from Compustat. Stock return and trading data are from CRSP. Analyst forecast data are from I/B/E/S. CEO turnover classifications, including forced versus voluntary departures, are from the dataset compiled by [Gentry et al. \(2021\)](#). We merge these datasets using GVKEY (Compustat–ExecuComp link) and CUSIP (I/B/E/S–Compustat link).

We exclude financial firms (SIC 6000–6999) and utilities (SIC 4900–4999), as is standard in corporate finance research. We require a minimum of 10 scored transcript segments per CEO-year for reliable individualism estimates. We drop CEO transition years (first year as CEO and year of departure) to avoid contamination from interim leadership arrangements. The final sample comprises approximately 9,700 CEO-year observations covering 1,078 firms and 2,096 CEOs.

3.2 Measuring CEO Individualism

3.2.1 Conceptual Framework

We measure CEO individualism along three dimensions derived from the cultural psychology literature on individualism–collectivism ([Hofstede, 1980](#); [Triandis, 1995](#)):

- **D1—Self-Referential Leadership:** The extent to which the CEO uses first-person singular pronouns, takes personal ownership of strategic decisions, and frames corporate actions as individual choices rather than organizational processes.
- **D2—Personal Communication Style:** The extent to which the CEO’s language is distinctive, expressive, and personality-revealing, as opposed to formulaic corporate discourse.
- **D3—Organizational Embedding (reverse):** The extent to which the CEO positions themselves as separate from—rather than embedded within—the organizational collective.

Each dimension is scored on a 1–5 Likert scale, where 1 represents highly collectivist and 5 represents highly individualist. Our composite individualism score is the equal-

weighted average of the three dimensions. Table 1 presents examples from actual earnings call transcripts to illustrate how the dimensions distinguish CEO communication styles.

3.2.2 LLM-Based Scoring Pipeline

We construct our measure using a two-stage LLM pipeline designed for scalability and reproducibility:

Stage 1: Teacher Labeling. We extract all CEO speech segments from quarterly earnings call transcripts, separating prepared remarks from Q&A responses. Each CEO’s speech within a call is divided into chunks of approximately 500 words. From this corpus of 463,646 chunks, we randomly sample 20,000 for labeling by Claude Sonnet (Anthropic), a frontier language model. Each chunk is scored along the three dimensions using a detailed rubric with anchoring examples. The rubric is validated on a set of 20 CEOs with known individualism profiles (e.g., Elon Musk, Steve Jobs as highly individualist; regional bank and utility CEOs as collectivist).

Stage 2: Student Model Fine-Tuning and Inference. We fine-tune a Qwen3-14B model (open-source, locally deployable) on the 20,000 labeled chunks using LoRA adaptation. The fine-tuned model achieves $R^2 = 0.752$ against the teacher labels on a held-out test set, with CEO-level ranking correlation (Spearman ρ) of 0.90. We deploy this model to score all 463,646 chunks. Chunk-level scores are aggregated to the CEO-quarter and CEO-year level using word-count-weighted means.

3.2.3 Validation

We validate the measure along several dimensions:

Internal Consistency. Split-half reliability, computed by randomly dividing each CEO’s chunks into two halves and correlating the resulting scores, yields a Spearman-Brown corrected coefficient of 0.93–0.97 depending on the minimum chunk threshold.

Temporal Stability. Correlating scores from the first and second halves of each CEO’s tenure yields $r = 0.82$, confirming that the measure captures a stable trait rather than transient mood.

Convergent Validity. The individualism score correlates at $r = 0.62$ with the first-person pronoun ratio ($I/(I + We)$), the leading text-based proxy for CEO narcissism (Chatterjee and Hambrick, 2007). However, in horse-race regressions, the two measures enter with opposite signs on capital expenditure, confirming they capture related but distinct constructs.

Discriminant Validity. The score is orthogonal to managerial overconfidence as measured by the Holder67 indicator ($r = -0.07$) and to corporate political ideology as measured by PAC CFscores ($r = 0.01$).

Independence from CEO Power. The individualism score survives controls for CEO tenure, CEO-chair duality, board independence, and CEO pay-performance sensitivity (Delta), indicating it captures personality rather than structural power.

Unexplained Variance. In a multivariate regression of individualism on all observable CEO and firm characteristics (age, tenure, gender, duality, ownership, compensation, firm size, Tobin's q , ROA, leverage, R&D intensity, firm age, board independence), only 5.6% of the variance is explained, confirming that the measure predominantly reflects unobservable personality.

3.3 Empirical Specifications

3.3.1 Performance-Turnover Sensitivity

Our primary specification examines whether CEO individualism modulates the sensitivity of forced turnover to firm performance:

$$\text{ForcedTurnover}_{i,t+1} = \alpha + \beta_1 \text{Indiv}_{i,t} + \beta_2 \text{AbRet}_{i,t} + \beta_3 \text{Indiv}_{i,t} \times \text{AbRet}_{i,t} + \gamma' X_{i,t} + \delta_{j \times t} + \varepsilon_{i,t} \quad (1)$$

where $\text{ForcedTurnover}_{i,t+1}$ is an indicator for CEO i being forcibly dismissed in fiscal year $t + 1$; $\text{Indiv}_{i,t}$ is the standardized individualism score; $\text{AbRet}_{i,t}$ is the standardized value-weighted abnormal stock return; $X_{i,t}$ is a vector of controls (firm size, leverage, CEO tenure, CEO age, gender); and $\delta_{j \times t}$ are industry (SIC2) $\times$ year fixed effects. Standard

errors are clustered by firm.

The coefficient of interest is β_3 . A negative β_3 indicates that individualism *increases* performance-turnover sensitivity: good performance reduces firing probability more for individualist CEOs, while bad performance increases it more.

We estimate Equation (1) using both linear probability models (LPM) and logit. The LPM facilitates interpretation of interaction terms, while logit better accommodates the rare-event nature of forced turnover (base rate $\approx 1.5\%$).

3.3.2 Market Microstructure

To test Hypothesis 2, we estimate:

$$Y_{i,t} = \alpha + \beta_1 \text{Indiv}_{i,t} + \gamma' X_{i,t} + \delta_{j \times t} + \varepsilon_{i,t} \quad (2)$$

where $Y_{i,t}$ is one of: log analyst coverage, log effective spread, log Amihud illiquidity, forecast dispersion, or absolute cumulative abnormal return around earnings announcements. Controls and fixed effects are as in Equation (1).

4 Empirical Results

4.1 Summary Statistics

Table 3 reports summary statistics. The mean CEO individualism score is 2.99 (SD = 0.28) on a 1–5 scale. Of the three dimensions, D3 (organizational embedding) has the highest mean (3.35) and D1 (self-referential leadership) the lowest (2.70). The forced turnover rate is 1.5% per year (154 events). Average analyst coverage is 11.6. The sample covers 1,078 firms and 2,096 CEOs after applying filters.

4.2 Measurement Validation

Table 4 summarizes the validation evidence.

Reliability. Split-half reliability (Spearman-Brown corrected), computed by randomly dividing each CEO's scored chunks into two halves and correlating the resulting individualism scores, is 0.927 for CEOs with at least 10 chunks, rising to 0.970 for CEOs with at least 80 chunks. Temporal stability (first-half versus second-half of each CEO's tenure) is $r = 0.82$.

Convergent validity. The individualism score correlates at $r = 0.62$ with the first-person pronoun ratio $I/(I + We)$, the standard text-based narcissism proxy from Chatterjee and Hambrick (2007). In CEO-level cross sections, the pronoun ratio for the highest individualism quintile is 0.244, versus 0.136 for the lowest. In horse-race regressions predicting R&D and capital expenditure, both measures are jointly significant but enter with *opposite signs* on capex: individualism predicts higher capex while the pronoun ratio predicts lower capex. This confirms the two measures capture related but distinct constructs.

Discriminant validity. The individualism score is orthogonal to managerial overconfidence as measured by the Holder67 indicator ($r = -0.07$, $p = 0.03$). In horse-race regressions, overconfidence predicts higher R&D while individualism predicts lower R&D—opposite directions. The score is also uncorrelated with corporate political ideology ($r = 0.01$ with PAC CFscores).

Independence from CEO power. We progressively add CEO tenure, CEO-chair duality, board independence, and CEO Delta (pay-performance sensitivity) to regressions of analyst coverage on individualism. The individualism coefficient declines from +0.060 ($t = 4.85$) to +0.054 ($t = 4.09$)—a reduction of only 10%—confirming that the measure captures personality rather than structural power.

Variance decomposition. The intraclass correlation coefficient (ICC) for CEO identity is 0.765, versus 0.586 for firm identity, with an incremental CEO effect of 0.179. In multi-CEO firms, R^2 from CEO dummies is 0.815, versus 0.552 from firm dummies alone—CEO identity explains 26.4% of variance beyond the firm. In succession pairs (888 consecutive CEO pairs at the same firm, each with at least 2 years), the correlation between predecessor and successor individualism is 0.44, and within-firm succession

$|\Delta \text{indiv}|$ is 27% smaller than random-pair $|\Delta \text{indiv}|$, indicating some CEO-firm matching but substantial CEO-specific variation. A multivariate regression of individualism on all observable CEO and firm characteristics yields $R^2 = 0.056$; 94.4% of the variance is unexplained.

Determinants of individualism. The strongest observable predictors of CEO individualism are: female (-0.139 , $t = -13.5$), CEO-chair duality ($+0.051$, $t = 7.5$), firm size ($+0.020$, $t = 7.7$), CEO ownership ($+0.004$, $t = 6.1$), firm age (-0.001 , $t = -5.9$), CEO tenure ($+0.003$, $t = 4.3$), board independence (-0.120 , $t = -3.9$), and log total compensation (-0.013 , $t = -2.9$). These patterns are consistent with CEOs who have more autonomy (duality, ownership, tenure, weak boards) exhibiting higher individualism.

4.3 Finding 1: Analyst Coverage and Market Microstructure

Table 7 reports cross-sectional regressions of information environment variables on CEO individualism with SIC2×year fixed effects and firm-clustered standard errors. A one-standard-deviation increase in individualism is associated with:

- 5.7% higher analyst coverage ($t = 4.62$, $p < 0.001$)
- 2.4% lower effective bid-ask spread ($t = -2.53$, $p = 0.011$)
- 5.5% lower Amihud illiquidity ($t = -3.50$, $p < 0.001$)
- No significant change in forecast dispersion ($t = -0.76$, $p = 0.45$)
- No significant change in absolute earnings announcement CAR ($t = 1.40$, $p = 0.16$)
- No significant change in institutional ownership ($t = -0.39$, $p = 0.69$)

The analyst coverage result is robust across specifications: it holds after adding R&D and capex controls ($t = 6.15$), using QA-only individualism scores ($t = 4.67$), dropping technology firms ($t = 5.13$), and including all CEO power controls ($t = 4.09$). All three

dimensions (D1, D2, D3) individually predict coverage. The result holds in firm fixed effects specifications ($t = 5.15$), though we note the caveat (discussed in Section 4.8) that firm FE identification is partially driven by within-CEO score variation rather than CEO switches.

4.4 Finding 2: Performance-Turnover Sensitivity

Table 5 reports linear probability models and Table 6 reports logit models of forced CEO turnover in year $t + 1$ on CEO individualism and firm performance in year t .

Main effect. The coefficient on standardized individualism is -0.0040 ($t = -2.85$, $p = 0.004$), indicating that, unconditionally, individualist CEOs are *less* likely to be forcibly dismissed in the following year. This is a reduction of 0.40 percentage points from a base rate of 1.53%, or 26% in relative terms.

Interaction with performance. In the specification including the interaction of individualism with standardized abnormal stock returns:

- $\hat{\beta}_1$ (individualism main effect) = -0.0038 ($t = -2.76$)
- $\hat{\beta}_2$ (abnormal return main effect) = $+0.0021$ ($t = 1.24$)
- $\hat{\beta}_3$ (interaction) = -0.0047 ($t = -2.93$, $p = 0.003$)

The total marginal effect of individualism on forced turnover depends on performance:

- At +1 SD return (good year): $-0.0038 + (-0.0047)(+1) = -0.0085$ (−56% relative)
- At mean return: -0.0038 (−25% relative)
- At −1 SD return (bad year): $-0.0038 + (-0.0047)(-1) = +0.0009$ (protection eliminated)
- At −2 SD return: $+0.0056$ (individualist CEOs face higher dismissal risk)

In the binary specification, the interaction of individualism with a negative-abnormal-return indicator is +0.0053 ($t = 2.07$, $p = 0.039$): the protective effect of individualism falls from -0.0067 per SD in good years to just -0.0014 in bad years.

Logit confirmation. In logit models, the interaction is -0.286 ($z = -3.67$, $p = 0.0002$), with a marginal effect of -0.0042 . The stronger significance in logit reflects its better fit for rare events.

2×2 turnover rates. The forced turnover rate by individualism (median split) and performance (positive/negative abnormal return):

	Positive AbRet	Negative AbRet
Low Individualism	2.11%	1.64%
High Individualism	1.04%	1.32%

Individualist CEOs have the lowest turnover rate (1.04%) after good performance and see it rise to 1.32% after poor performance. Collectivist CEOs' rate is relatively stable (2.11% to 1.64%).

Any turnover. The main effect of individualism on any CEO turnover (forced + voluntary) is also significant: +0.014 ($t = 4.38$, $p < 0.001$) for the lead specification.

4.5 Finding 3: Corporate Investment

In cross-sectional regressions with SIC2×year fixed effects, a one-standard-deviation increase in individualism is associated with:

- Lower R&D/assets: -0.0041 ($t = -3.27$, $p = 0.001$) using QA-only scores
- Higher capex/assets: $+0.0015$ ($t = 2.11$, $p = 0.035$) using overall scores

These effects do not survive firm fixed effects, suggesting they reflect CEO-firm sorting rather than within-firm causal effects. Founder CEOs exhibit independently higher R&D ($+0.023$, $t = 3.0$), and the founder×individualism interaction is significant for capex ($+0.006$, $t = 2.6$) but not for R&D.

4.6 Finding 4: Secular Decline

Mean CEO individualism declined from 3.13 in 2007 to 2.95 in 2024, a drop of 0.18 points (approximately 0.6 SD). The linear trend is -0.006 per year ($t = -14.1$). Decomposition into within-CEO and between-CEO components shows:

- Within-CEO trend: -0.0003 /year ($t = -1.46$, not significant)—individual CEOs do not become more or less individualist over time.
- Between-CEO (cohort) trend: -0.0072 /year ($t = -8.92$)—newly appointed CEOs are systematically more collectivist than those they replace.

The decline is most pronounced in D3 (organizational embedding, -0.010 /year) and D2 (personal style, -0.006 /year), and weakest in D1 (self-reference, -0.003 /year).

4.7 Additional Findings

Founder CEOs. CEOs whose ExecuComp title contains “Founder” score 0.156 higher ($t = 6.98$). In the highest individualism quintile, 11.2% are founders versus 2.7% in the lowest.

CEO Delta. Individualism is positively associated with CEO Delta (pay-performance sensitivity): the highest individualism quintile has mean Delta of \$5,705K versus \$552K in the lowest. However, individualism predicts analyst coverage independently of Delta (both significant in the same regression).

Risk-taking. Individualism is not significantly associated with stock return volatility, idiosyncratic volatility, or leverage after controlling for firm characteristics. Cash flow volatility is marginally negatively associated ($t = -1.98$).

Performance. Individualism has no main effect on ROA, Tobin’s q , or stock returns. The interaction of individualism with Delta on abnormal stock returns is marginally significant ($+0.016$, $t = 2.01$, $p = 0.044$).

4.8 Identification Limitations

We note several limitations regarding causal inference:

Firm fixed effects. The analyst coverage result is significant in firm FE specifications ($t = 5.15$). However, diagnostic tests reveal that this effect is also present in single-CEO firms ($t = 6.41$)—where the only within-firm variation comes from year-to-year noise in the same CEO’s score. This suggests the firm FE result is partially driven by correlated noise rather than CEO switches.

CEO transition event study. We identify 1,108 CEO transitions in our panel. Regressing the change in analyst coverage on the change in individualism yields no significant effect ($t \approx 0$). Using CEO-tenure-mean scores (888 succession pairs, each CEO observed for at least 2 years), the association is marginal ($r = 0.066$, $p = 0.11$).

Interpretation. The cross-sectional associations we document are robust to extensive controls (industry×year FE, firm characteristics, CEO power proxies) but could reflect endogenous CEO-firm matching rather than causal CEO effects. We present these as empirical regularities that invite further investigation with stronger identification strategies.

5 Conclusion

This paper introduces a novel LLM-based measure of CEO individualism from earnings call transcripts and documents its empirical associations with governance, market, and investment outcomes. The measure is psychometrically sound (split-half reliability 0.93), captures a dimension distinct from overconfidence and CEO power, and is largely driven by unobservable CEO personality (94% of variance unexplained by observables).

We document four sets of findings. First, individualist CEOs are associated with more analyst coverage and greater stock market liquidity, without improved forecast accuracy—an “attention without information” pattern. Second, CEO individualism increases the sensitivity of forced turnover to firm performance: individualist CEOs are better protected after good performance but more exposed after poor performance. Third,

individualism is cross-sectionally associated with a preference for tangible investment (capital expenditure) over intangible investment (R&D). Fourth, CEO individualism has declined steadily over two decades, driven entirely by the entry of more collectivist cohorts.

We emphasize that these findings are presented as documented associations rather than causal claims. Cross-sectional associations do not survive CEO transition event studies, and firm fixed effects are partially confounded by within-CEO measurement noise. The performance-turnover interaction, while robust and highly significant in logit specifications, is identified from cross-sectional variation. Establishing causality will require stronger identification strategies, such as instrumental variables based on CEO pre-determined characteristics or quasi-experimental CEO assignment.

Our primary contribution is methodological: we demonstrate that LLMs can extract meaningful, stable, and predictive personality measures from corporate communication at scale. The individualism score complements existing CEO characteristic measures and is available at quarterly frequency for any CEO with earnings call transcripts. We make the scoring rubric and pipeline code available to facilitate replication and extension to other personality dimensions.

References

- Ahern, Kenneth R, Daniele Daminelli, and Cesare Fracassi (2015) “Lost in translation? The effect of cultural values on mergers around the world,” *Journal of Financial Economics*, 117 (1), 165–189.
- Amihud, Yakov (2002) “Illiquidity and stock returns: Cross-section and time-series effects,” *Journal of Financial Markets*, 5 (1), 31–56.
- Benmelech, Efraim and Carola Frydman (2015) “Military CEOs,” *Journal of Financial Economics*, 117 (1), 43–59.
- Bertrand, Marianne and Antoinette Schoar (2003) “Managing with style: The effect of managers on firm policies,” *The Quarterly Journal of Economics*, 118 (4), 1169–1208.
- Chatterjee, Arijit and Donald C Hambrick (2007) “It’s all about me: Narcissistic chief executive officers and their effects on company strategy and performance,” *Administrative Science Quarterly*, 52 (3), 351–386.
- Custódio, Cláudia, Miguel A Ferreira, and Pedro Matos (2013) “Generalists versus specialists: Lifetime work experience and chief executive officer pay,” *Journal of Financial Economics*, 108 (2), 471–492.
- Gentry, Richard J, Joseph S Harrison, Timothy J Quigley, and Steven Boivie (2021) “A database of CEO turnover and dismissal in S&P 1500 firms, 2000–2018,” *Strategic Management Journal*, 42 (5), 968–991.
- Gorodnichenko, Yuriy and Gérard Roland (2011) “Which dimensions of culture matter for long-run growth?” *American Economic Review*, 101 (3), 492–498.
- Hermalin, Benjamin E and Michael S Weisbach (1998) “Endogenously chosen boards of directors and their monitoring of the CEO,” *American Economic Review*, 88 (1), 96–118.
- Hofstede, Geert (1980) *Culture’s Consequences: International Differences in Work-Related Values*: Sage Publications.
- Jenter, Dirk and Katharina Lewellen (2015) “CEO preferences and acquisitions,” *The Journal of Finance*, 70 (6), 2813–2852.
- Li, Kai, Dale Griffin, Heng Yue, and Longkai Zhao (2013) “Does individualism influence stock price crash risk?” *Working paper*.
- Loughran, Tim and Bill McDonald (2011) “When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks,” *The Journal of Finance*, 66 (1), 35–65.
- Malmendier, Ulrike and Geoffrey Tate (2005) “CEO overconfidence and corporate investment,” *The Journal of Finance*, 60 (6), 2661–2700.

——— (2009) “Superstar CEOs,” *The Quarterly Journal of Economics*, 124 (4), 1593–1638.

Markus, Hazel Rose and Shinobu Kitayama (1991) “Culture and the self: Implications for cognition, emotion, and motivation,” *Psychological Review*, 98 (2), 224–253.

Parrino, Robert, Richard W Sias, and Laura T Starks (2003) “Voting with their feet: Institutional ownership changes around forced CEO turnover,” *Journal of Financial Economics*, 68 (1), 3–46.

Triandis, Harry C (1995) *Individualism and Collectivism*: Westview Press.

Weisbach, Michael S (1988) “Outside directors and CEO turnover,” *Journal of Financial Economics*, 20, 431–460.

Table 1: Illustrative Examples of Individualism Scoring from Earnings Call Q&A

Dim.	Score	Q&A Excerpt
<i>Panel A: D1—Self-Referential Leadership</i>		
D1 = 5	HOLX, 2023 Q3	“Let’s be really clear. I’m at Hologic through the end of my useful life in terms of what I’ve worked for, built for. And as a reminder to everybody, I am personally a top 15 shareholder in Hologic. So this is my day job. This is my passion. This is my love.” —Stephen MacMillan, CEO of Hologic, answering a question about his commitment. CEO frames the company as a personal project.
D1 = 1	BKH, 2015 Q1	“We earned \$1.07 per share compared to \$1.08 per share in the first quarter of 2014. Overall an excellent result considering the reduction in heating degree days and the reduction in oil and gas prices compared to last year.” —David Emery, CEO of Black Hills Corp. CEO is invisible; the organization reports its own results.
<i>Panel B: D2—Personal Communication Style</i>		
D2 = 5	TXRH, 2013 Q1	“I’d refer back to the kicking ass comment. We’re always looking for other brands. That might be our second avenue... They’ve been on TV a lot lately and our sales have been pretty good. So maybe we’d like to thank them for helping us.” —Kent Taylor, CEO of Texas Roadhouse. Informal, humorous, colloquial—“kicking ass” in an earnings call.
D2 = 1	FLS, 2019 Q4	“Turning to fourth quarter bookings by geography, Europe delivered 17% growth with Asia-Pacific and Latin America contributing growth of 11% and 8%, respectively. North America and the Middle East and Africa were both down roughly 9%.” —Scott Rowe, CEO of Flowserve. Pure data recitation, zero personality.
<i>Panel C: D3—Organizational Embedding</i>		
D3 = 5	CLF, 2024 Q2	“Look, the President of the United States said loud and clear that U.S. Steel will be American-owned and American-operated. That does include anybody from Japan, particularly when the Head Executive said that he likes to wait for the election because after the election the USW will have no leverage. Think about that.” —Lourenco Goncalves, CEO of Cleveland-Cliffs. CEO speaks as an autonomous force commenting on national policy, not as a company representative.
D3 = 1	CVBF, 2019 Q1	“Non-interest expense for the first quarter was \$51.6 million compared with \$60.8 million for the fourth quarter of 2018. The first quarter of 2019 included \$3.1 million in acquisition expenses compared with \$8.5 million for the prior quarter.” —Christopher Myers, CEO of CVB Financial. CEO functions purely as an organizational spokesperson reading financial data.

Notes: All excerpts are from the Q&A session of quarterly earnings calls, where CEO responses are largely spontaneous (as opposed to prepared remarks, which are typically drafted by investor relations teams). Each panel contrasts a score of 5 (highly individualist) with 1 (highly collectivist) on one dimension. Panel A illustrates how D1 distinguishes personal ownership of the narrative (“I am... this is my passion”) from invisible organizational report-

Table 2: Determinants of CEO Individualism

Variable	Bivariate		Multivariate OLS	
	Pearson r	p -value	Coefficient	t -stat
<i>CEO Characteristics</i>				
Female	−0.151	< 0.001	−0.139	−13.5***
CEO-Chair Duality	+0.127	< 0.001	+0.051	+7.5***
CEO Ownership (%)	+0.134	< 0.001	+0.004	+6.1***
CEO Tenure (years)	+0.080	< 0.001	+0.003	+4.3***
CEO Age	+0.045	< 0.001	−0.000	−0.0
Log Total Compensation	−0.013	0.203	−0.013	−2.9***
External Hire	+0.111	< 0.001		
<i>Firm Characteristics</i>				
Firm Size (log assets)	+0.047	< 0.001	+0.020	+7.7***
Firm Age	−0.047	< 0.001	−0.001	−5.9***
Tobin's q	+0.018	0.075	+0.004	+2.8***
Leverage	+0.031	0.002	+0.045	+2.6***
ROA	−0.002	0.863	−0.035	−1.4
R&D / Assets	−0.002	0.856	−0.000	−0.0
<i>Governance</i>				
Board Independence	−0.065	< 0.001	−0.120	−3.9***
N			8,981	
R^2			0.056	
Unexplained variance			94.4%	

Notes: The left panel reports bivariate Pearson correlations between CEO individualism and each variable. The right panel reports coefficients from an OLS regression of CEO individualism on all listed variables simultaneously, with heteroskedasticity-robust standard errors. The dependent variable is the composite individualism score (mean of D1, D2, D3). The low R^2 of 0.056 indicates that observable CEO and firm characteristics jointly explain only 5.6% of the variation in individualism; the remaining 94.4% reflects unobservable personality. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Table 3: Summary Statistics

Variable	N	Mean	SD	P25	Median	P75
<i>Panel A: CEO Individualism Score</i>						
Individualism (composite)	9,722	2.99	0.28	2.80	2.98	3.17
D1: Self-Referential Leadership	9,722	2.70	0.24			
D2: Personal Communication Style	9,722	2.90	0.32			
D3: Organizational Embedding	9,722	3.35	0.25			
<i>Panel B: Turnover</i>						
Forced Turnover	9,722	0.015	0.123	0	0	0
Any CEO Turnover	9,722	0.101	0.301	0	0	0
<i>Panel C: Firm Characteristics</i>						
Firm Size (log assets)	9,722					
Leverage	9,722					
Tobin's q	9,722					
ROA	9,722					
Abnormal Return (VW)	9,722					
<i>Panel D: CEO Characteristics</i>						
CEO Age	9,722					
CEO Tenure	9,722					
CEO-Chair Duality	9,348					
Female CEO	9,722	0.053				
<i>Panel E: Information Environment</i>						
Analyst Coverage	9,632	11.6			10	
Forecast Dispersion	9,433					
Effective Spread	9,716					
Amihud Illiquidity	9,716					

Notes: This table reports summary statistics for the main analysis sample. The sample covers 2005–2024, excludes financial firms (SIC 6000–6999) and utilities (SIC 4900–4999), requires at least 10 scored transcript segments per CEO-year, and drops CEO transition years. [TBD: fill in remaining cells from data.]

Table 4: Measurement Validation

Validation Type	Metric	Value	<i>N</i>
<i>Panel A: Reliability</i>			
Split-half (min 10 chunks)	Spearman-Brown	0.927	3,359 CEOs
Split-half (min 20 chunks)	Spearman-Brown	0.943	3,126 CEOs
Split-half (min 40 chunks)	Spearman-Brown	0.959	2,690 CEOs
Temporal stability (1st vs 2nd half)	Pearson <i>r</i>	0.818	3,126 CEOs
<i>Panel B: Convergent Validity</i>			
vs. First-person pronoun ratio	Pearson <i>r</i>	0.617	3,359 CEOs
vs. Founder CEO indicator	<i>t</i> -test	<i>t</i> = 6.98	3,359 CEOs
<i>Panel C: Discriminant Validity</i>			
vs. Overconfidence (Holder67)	Pearson <i>r</i>	−0.070	1,009 CEOs
vs. PAC Ideology (CFscore)	Pearson <i>r</i>	0.009	1,221 CEOs
<i>Panel D: Variance Decomposition</i>			
ICC (CEO identity)		0.765	
ICC (Firm identity)		0.586	
CEO increment beyond firm		0.179	
<i>R</i> ² from all observables		0.056	8,981 obs

Notes: Panel A reports split-half reliability computed by randomly dividing each CEO's transcript chunks into two halves (averaged over 10 random splits). Panel B reports convergent validity with the pronoun ratio $I/(I + We)$ from Chatterjee and Hambrick (2007). Panel C reports discriminant validity against the Holder67 overconfidence indicator and corporate PAC ideology scores. Panel D reports intraclass correlation coefficients and R^2 from regressing individualism on all observable CEO and firm characteristics.

Table 5: CEO Individualism and Performance-Turnover Sensitivity

	(1) Baseline	(2) + AbRet	(3) + Interaction	(4) Binary
<i>DV: Forced Turnover_{t+1}</i>				
<i>z_indiv</i>	−0.0040*** (0.0014)	−0.0040*** (0.0014)	−0.0038*** (0.0014)	−0.0067*** (0.0020)
<i>z_abret</i>		0.0021 (0.0017)	0.0021 (0.0017)	
<i>z_indiv</i> × <i>z_abret</i>			−0.0047*** (0.0016)	
$\mathbb{I}[AbRet < 0]$				−0.0010 (0.0027)
<i>z_indiv</i> × $\mathbb{I}[AbRet < 0]$				0.0053** (0.0026)
Controls	Yes	Yes	Yes	Yes
SIC2 × Year FE	Yes	Yes	Yes	Yes
Cluster (Firm)	Yes	Yes	Yes	Yes
<i>N</i>	8,643	8,643	8,643	8,643
<i>R</i> ²	0.110	0.110	0.112	0.111

Notes: This table reports LPM estimates of Equation (1). The dependent variable is an indicator for forced CEO turnover in fiscal year $t + 1$. *z_indiv* is the standardized CEO individualism score in year t . *z_abret* is the standardized value-weighted abnormal stock return in year t . Controls include firm size (log assets), leverage, CEO tenure, CEO age, and a female indicator. Industry (SIC2) × year fixed effects are included. Standard errors (in parentheses) are clustered by firm. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Table 6: CEO Individualism and Forced Turnover: Logit Estimates

	(1) Baseline	(2) + AbRet Interaction	(3) + ROA Interaction
z_indiv	-0.260***	-0.228** ($z = -2.44$)	-0.280*** ($z = -2.90$)
z_abret		0.058 ($z = 0.63$)	
$z_indiv \times z_abret$		-0.286*** ($z = -3.67$)	
z_roa			-0.306*** ($z = -4.02$)
$z_indiv \times z_roa$			-0.055 ($z = -0.77$)
Controls	Yes	Yes	Yes
Pseudo R^2	0.019	0.031	0.029
N	8,643	8,643	8,643

Notes: This table reports logit estimates of forced turnover in year $t + 1$ on CEO individualism and firm performance in year t . Controls include firm size, leverage, CEO tenure, CEO age, and a female indicator. z -statistics are reported. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Table 7: CEO Individualism and Information Environment

	(1) Coverage (log)	(2) Eff. Spread (log)	(3) Amihud (log)	(4) Disp.	(5) CAR [-1, +1]
z_indiv	0.057*** (0.012)	-0.024** (0.010)	-0.055*** (0.016)	-0.005 (0.006)	0.001 (0.001)
t -stat	4.62	-2.53	-3.50	-0.76	1.40
Controls	Yes	Yes	Yes	Yes	Yes
SIC2 $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Cluster (Firm)	Yes	Yes	Yes	Yes	Yes
N	9,610	9,694	9,694	9,411	8,750

Notes: This table reports OLS estimates of Equation (2). Each column uses a different dependent variable. Coverage is the log number of analysts issuing annual EPS forecasts (I/B/E/S). Effective spread and Amihud illiquidity are annual averages of daily values from CRSP. Dispersion is the standard deviation of analyst forecasts scaled by the absolute mean forecast. |CAR| is the absolute cumulative abnormal return in the [-1, +1] window around earnings announcements. Controls include firm size, leverage, Tobin's q , ROA, CEO age, and a female indicator. Standard errors clustered by firm. ***, **, * denote significance at 1%, 5%, and 10%.

Technical Appendix

A. LLM Scoring Pipeline Details

A.1 Transcript Processing

Earnings call transcripts are sourced from S&P Capital IQ, covering 2005–2024. Each transcript is parsed into speaker turns, and we identify the CEO using name matching against ExecuComp records. Transcripts are segmented into “prepared remarks” (CEO’s opening statement) and “Q&A” (CEO’s responses to analyst questions) based on operator cues (e.g., “We will now begin the question-and-answer session”).

CEO speech within each section is divided into chunks of approximately 500 words, preserving sentence boundaries. This chunking balances two objectives: providing sufficient context for reliable scoring and maintaining manageable input length for the language model. The resulting corpus contains 463,646 chunks across 61,581 CEO-quarter observations.

A.2 Teacher Model Labeling

We use Claude Sonnet (Anthropic, 2024) as the teacher model. Each chunk is scored using the following detailed rubric, provided as a system prompt:

D1: Self-Referential Leadership

- 1 = Company/team is always the subject. “We delivered strong results.” CEO is invisible.
- 2 = Mostly “we”, occasional “I think” as hedge, not as personal claim.
- 3 = Mix of “I” and “we”. CEO occasionally states personal views.
- 4 = CEO frequently narrates as personal protagonist. “I decided to restructure.”
- 5 = CEO is the hero of the story. Company achievements = personal achievements.

D2: Personal Communication Style

- 1 = Fully scripted/institutional. Reads prepared remarks by business segment. Zero anecdotes.

- 2 = Mostly formal, occasional colloquial expression. Data-driven, restrained.
- 3 = Mix of formal and personal. Some analogies, natural tone, stays on topic.
- 4 = Clearly personal. Uses stories/humor/emotion as arguments. Occasionally tangential.
- 5 = Extremely informal. Stream-of-consciousness, jokes, strong emotions, frequent tangents.

D3: Organizational Embedding

- 1 = Actively defers questions to other executives. “I’ll let our CFO address that.”
- 2 = Often references colleagues/processes. Shares speaking time.
- 3 = Answers most questions but frames with “the team believes”, “our analysis shows”.
- 4 = Answers all questions personally with deep detail. Shows personal command.
- 5 = Completely dominates. Frames strategy as personal will. “I told the team we’re doing this.”

Scoring rules:

- Score only what is observed in the text. Do not use external knowledge about the CEO.
- Be consistent: same language patterns should always receive the same score.
- Most earnings call speech falls in the 2–4 range. Use 1 or 5 only for clear extremes.

The model outputs a JSON object { ‘D1’: <int>, ‘D2’: <int>, ‘D3’: <int>} with temperature set to 0 for reproducibility. We label 20,000 randomly sampled chunks using the Anthropic Batch API. Quality is assessed against a validation set of 20 CEOs with known individualism profiles (e.g., Elon Musk, Steve Wynn as highly individualist; regional bank and utility CEOs as collectivist).

A.3 Student Model Fine-Tuning

We fine-tune Qwen3-14B (open-source, 14 billion parameters) using LoRA (Low-Rank Adaptation) on the 20,000 teacher-labeled chunks. Training details:

- Effective batch size: 8 (batch size 4, gradient accumulation 2)

- Learning rate: $2e-5$ with cosine schedule
- Training epochs: 4,000 steps
- Hardware: NVIDIA RTX 3090 Ti (24GB VRAM)
- Training time: approximately 5–6 hours

The fine-tuned model achieves $R^2 = 0.752$ on a held-out test set (exact agreement 80–86%, within ± 1 agreement 100%). CEO-level ranking correlation (Spearman ρ) between teacher and student scores is 0.90.

A.4 Aggregation

Chunk-level scores are aggregated to the CEO-quarter level using word-count-weighted means:

$$\text{Indiv}_{i,q} = \frac{\sum_{c \in C_{i,q}} w_c \cdot s_c}{\sum_{c \in C_{i,q}} w_c}$$

where $C_{i,q}$ is the set of chunks for CEO i in quarter q , w_c is word count, and s_c is the composite score $(D1 + D2 + D3)/3$. CEO-year scores are similarly aggregated from quarters, weighted by the number of chunks per quarter.

B. Variable Definitions

- **Forced Turnover:** Indicator for CEO being involuntarily dismissed, from [Gentry et al. \(2021\)](#). Excludes retirements, health-related departures, and M&A-related changes.
- **Abnormal Return (VW):** Annual stock return minus the value-weighted CRSP market return.
- **Analyst Coverage:** Number of analysts issuing annual EPS forecasts (I/B/E/S), measured at the last statistical period before fiscal year-end.
- **Effective Spread:** Annual average of daily effective bid-ask spread from CRSP, defined as $2 \times |P_{trade} - M|/M$ where M is the midpoint.

- **Amihud Illiquidity:** Annual average of daily $|r_t|/\text{DollarVolume}_t$ (Amihud, 2002).
- **Forecast Dispersion:** Standard deviation of analyst EPS forecasts divided by the absolute mean forecast.
- **CEO Delta:** Dollar change in CEO wealth per 1% change in stock price.
- **CEO Duality:** Indicator for CEO also serving as board chair.
- **Board Independence:** Fraction of independent directors on the board.